

GridBalance — Data Export — Gridbalance Master Results

Official Research Data Export — Total Records: 90

dataset	experiment	model	feature_set	strategy	split	metric	value	source_file	notes
SGCC	Baseline Evaluation	Logistic Regression	42 Features	Original	Validation	Precision	0.523810	REAL_SGCC_BASELINE_RESULTS.csv	No class weighting, median imputer
SGCC	Baseline Evaluation	Logistic Regression	42 Features	Original	Validation	Recall	0.040590	REAL_SGCC_BASELINE_RESULTS.csv	No class weighting, median imputer
SGCC	Baseline Evaluation	Logistic Regression	42 Features	Original	Validation	F1	0.075342	REAL_SGCC_BASELINE_RESULTS.csv	No class weighting, median imputer
SGCC	Baseline Evaluation	Logistic Regression	42 Features	Original	Validation	PR-AUC	0.259303	REAL_SGCC_BASELINE_RESULTS.csv	No class weighting, median imputer
SGCC	Baseline Evaluation	Logistic Regression	42 Features	Original	Validation	ROC-AUC	0.753086	REAL_SGCC_BASELINE_RESULTS.csv	No class weighting, median imputer
SGCC	Baseline Evaluation	Decision Tree	42 Features	Original	Validation	Precision	0.250426	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted tree
SGCC	Baseline Evaluation	Decision Tree	42 Features	Original	Validation	Recall	0.271218	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted tree
SGCC	Baseline Evaluation	Decision Tree	42 Features	Original	Validation	F1	0.260407	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted tree
SGCC	Baseline Evaluation	Decision Tree	42 Features	Original	Validation	PR-AUC	0.130076	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted tree
SGCC	Baseline Evaluation	Decision Tree	42 Features	Original	Validation	ROC-AUC	0.591369	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted tree
SGCC	Baseline Evaluation	Random Forest	42 Features	Original	Validation	Precision	0.644068	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted forest
SGCC	Baseline Evaluation	Random Forest	42 Features	Original	Validation	Recall	0.070111	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted forest
SGCC	Baseline Evaluation	Random Forest	42 Features	Original	Validation	F1	0.126456	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted forest
SGCC	Baseline Evaluation	Random Forest	42 Features	Original	Validation	PR-AUC	0.343214	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted forest

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SGCC	Baseline Evaluation	Random Forest	42 Features	Original	Validation	ROC-AUC	0.816903	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted forest
SGCC	Baseline Evaluation	XGBoost	42 Features	Original	Validation	Precision	0.556818	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted gradient boosting
SGCC	Baseline Evaluation	XGBoost	42 Features	Original	Validation	Recall	0.180812	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted gradient boosting
SGCC	Baseline Evaluation	XGBoost	42 Features	Original	Validation	F1	0.272981	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted gradient boosting
SGCC	Baseline Evaluation	XGBoost	42 Features	Original	Validation	PR-AUC	0.398049	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted gradient boosting
SGCC	Baseline Evaluation	XGBoost	42 Features	Original	Validation	ROC-AUC	0.826893	REAL_SGCC_BASELINE_RESULTS.csv	Unweighted gradient boosting
SGCC	Class Weighting	Logistic Regression	42 Features	Weighted	Validation	F1	0.285050	REAL_SGCC_BASELINE_RESULTS.csv	Class weighted
SGCC	Class Weighting	Decision Tree	42 Features	Weighted	Validation	F1	0.242764	REAL_SGCC_BASELINE_RESULTS.csv	Class weighted
SGCC	Class Weighting	Random Forest	42 Features	Weighted	Validation	F1	0.308497	REAL_SGCC_BASELINE_RESULTS.csv	Class weighted
SGCC	Class Weighting	XGBoost	42 Features	Weighted	Validation	F1	0.412698	REAL_SGCC_BASELINE_RESULTS.csv	Class weighted
SGCC	SMOTE Oversampling	Logistic Regression	42 Features	SMOTE	Validation	F1	0.289628	REAL_SGCC_BASELINE_RESULTS.csv	SMOTE inside pipeline
SGCC	SMOTE Oversampling	Decision Tree	42 Features	SMOTE	Validation	F1	0.264384	REAL_SGCC_BASELINE_RESULTS.csv	SMOTE inside pipeline
SGCC	SMOTE Oversampling	Random Forest	42 Features	SMOTE	Validation	F1	0.368473	REAL_SGCC_BASELINE_RESULTS.csv	SMOTE inside pipeline
SGCC	SMOTE Oversampling	XGBoost	42 Features	SMOTE	Validation	F1	0.411825	REAL_SGCC_BASELINE_RESULTS.csv	SMOTE inside pipeline
SGCC	Baseline Sealed Test	Logistic Regression	42 Features	Original	Sealed Test	F1	0.072664	REAL_SGCC_BASELINE_RESULTS.csv	Sealed test set

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SGCC	Baseline Sealed Test	Decision Tree	42 Features	Original	Sealed Test	F1	0.256591	REAL_SGCC_BASELINE_RESULTS.csv	Sealed test set
SGCC	Baseline Sealed Test	Random Forest	42 Features	Original	Sealed Test	F1	0.135091	REAL_SGCC_BASELINE_RESULTS.csv	Sealed test set
SGCC	Baseline Sealed Test	XGBoost	42 Features	Original	Sealed Test	F1	0.272981	REAL_SGCC_BASELINE_RESULTS.csv	Sealed test set
SGCC	Baseline Sealed Test	XGBoost	42 Features	Weighted	Sealed Test	F1	0.380222	REAL_SGCC_BASELINE_RESULTS.csv	Sealed test set
SGCC	Hyperparameter Tuning	Tuned XGBoost	42 Features	Weighted (Tuned)	5-Fold CV	F1	0.409000	SGCC_XGBOOST_TUNING_REPORT.md	CV Mean F1
SGCC	Hyperparameter Tuning	Tuned XGBoost	42 Features	Weighted (Tuned)	Validation	F1	0.457364	SGCC_XGBOOST_TUNING_RESULTS.csv	Threshold 0.50
SGCC	Hyperparameter Tuning	Tuned XGBoost	42 Features	Weighted (Tuned)	Validation	PR-AUC	0.422784	SGCC_XGBOOST_TUNING_RESULTS.csv	Threshold 0.50
SGCC	Hyperparameter Tuning	Tuned XGBoost	42 Features	Weighted (Tuned)	Validation	ROC-AUC	0.833437	SGCC_XGBOOST_TUNING_RESULTS.csv	Threshold 0.50
SGCC	Hyperparameter Tuning	Tuned XGBoost	42 Features	Weighted (Tuned)	Sealed Test	Precision	0.344737	SGCC_XGBOOST_TUNING_RESULTS.csv	Threshold 0.50
SGCC	Hyperparameter Tuning	Tuned XGBoost	42 Features	Weighted (Tuned)	Sealed Test	Recall	0.483395	SGCC_XGBOOST_TUNING_RESULTS.csv	Threshold 0.50
SGCC	Hyperparameter Tuning	Tuned XGBoost	42 Features	Weighted (Tuned)	Sealed Test	F1	0.402458	SGCC_XGBOOST_TUNING_RESULTS.csv	Threshold 0.50
SGCC	Hyperparameter Tuning	Tuned XGBoost	42 Features	Weighted (Tuned)	Sealed Test	PR-AUC	0.375018	SGCC_XGBOOST_TUNING_RESULTS.csv	Threshold 0.50
SGCC	Hyperparameter Tuning	Tuned XGBoost	42 Features	Weighted (Tuned)	Sealed Test	ROC-AUC	0.826008	SGCC_XGBOOST_TUNING_RESULTS.csv	Threshold 0.50
SGCC	Feature Selection	Full Model	42 Features	Weighted	5-Fold CV	CV F1 Mean	0.409334	SGCC_FEATURE_SET_COMPARISON.csv	5-fold CV on train/val
SGCC	Feature Selection	Full Model	42 Features	Weighted	Sealed Test	Test F1	0.416107	SGCC_FEATURE_SET_COMPARISON.csv	Sealed test evaluation

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SGCC	Feature Selection	Candidate Model	18 Features	Weighted	5-Fold CV	CV F1 Mean	0.402925	SGCC_FEATURE_SET_COMPARISON.csv	5-fold CV on train/val
SGCC	Feature Selection	Candidate Model	18 Features	Weighted	Sealed Test	Test Precision	0.318527	SGCC_FEATURE_SET_COMPARISON.csv	Sealed test evaluation
SGCC	Feature Selection	Candidate Model	18 Features	Weighted	Sealed Test	Test Recall	0.542435	SGCC_FEATURE_SET_COMPARISON.csv	Sealed test evaluation
SGCC	Feature Selection	Candidate Model	18 Features	Weighted	Sealed Test	Test F1	0.401365	SGCC_FEATURE_SET_COMPARISON.csv	Sealed test evaluation
SGCC	Feature Selection	Candidate Model	18 Features	Weighted	Sealed Test	Test PR-AUC	0.404908	SGCC_FEATURE_SET_COMPARISON.csv	Sealed test evaluation
SGCC	Feature Selection	Candidate Model	18 Features	Weighted	Sealed Test	Test ROC-AUC	0.834699	SGCC_FEATURE_SET_COMPARISON.csv	Sealed test evaluation
SGCC	Feature Selection	Stability Model	14 Features	Weighted	5-Fold CV	CV F1 Mean	0.369514	SGCC_FEATURE_SET_COMPARISON.csv	Folds selected >= 80%
SGCC	Feature Selection	Stability Model	14 Features	Weighted	Sealed Test	Test F1	0.364676	SGCC_FEATURE_SET_COMPARISON.csv	Folds selected >= 80%
SGCC	Feature Ablation	No Missingness	37 Features	Weighted	Validation	F1	0.383260	SGCC_MISSINGNESS_ABILATION.csv	All 5 missingness features removed
SGCC	Feature Ablation	No Missingness	37 Features	Weighted	Sealed Test	Precision	0.289062	SGCC_MISSINGNESS_ABILATION.csv	All 5 missingness features removed
SGCC	Feature Ablation	No Missingness	37 Features	Weighted	Sealed Test	Recall	0.409594	SGCC_MISSINGNESS_ABILATION.csv	All 5 missingness features removed
SGCC	Feature Ablation	No Missingness	37 Features	Weighted	Sealed Test	F1	0.338931	SGCC_MISSINGNESS_ABILATION.csv	All 5 missingness features removed
SGCC	Feature Ablation	No Missingness	37 Features	Weighted	Sealed Test	PR-AUC	0.290324	SGCC_MISSINGNESS_ABILATION.csv	All 5 missingness features removed
SGCC	Feature Ablation	No Missingness	37 Features	Weighted	Sealed Test	ROC-AUC	0.769105	SGCC_MISSINGNESS_ABILATION.csv	All 5 missingness features removed
SGCC	Robustness Noise	42-Feature Model	42 Features	Additive 20%	Sealed Test	F1	0.414229	SGCC_ROBUSTNESS_SUMMARY.csv	20% Additive Gaussian noise

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SGCC	Robustness Noise	42-Feature Model	42 Features	Additive 20%	Sealed Test	Flip Rate	0.051770	SGCC_ROBUSTNESS_SUMMARY.csv	Prediction flip rate under 20% noise
SGCC	Robustness Noise	18-Feature Model	18 Features	Additive 20%	Sealed Test	F1	0.407982	SGCC_ROBUSTNESS_SUMMARY.csv	20% Additive Gaussian noise
SGCC	Robustness Noise	18-Feature Model	18 Features	Additive 20%	Sealed Test	Flip Rate	0.059638	SGCC_ROBUSTNESS_SUMMARY.csv	Prediction flip rate under 20% noise
SGCC	Robustness Data Loss	42-Feature Model	42 Features	Long Burst 30%	Sealed Test	F1	0.348200	SGCC_MISSINGNESS_ROBUSTNESS_RESULTS.csv	30% long burst missingness
SGCC	Robustness Data Loss	18-Feature Model	18 Features	Long Burst 30%	Sealed Test	F1	0.340100	SGCC_MISSINGNESS_ROBUSTNESS_RESULTS.csv	30% long burst missingness
UCI	Forecasting Comparison	Linear Regression	29 Features	Default	Validation	MAE	3994.49	UCI_MODEL_COMPARISON.csv	Baseline Linear Model
UCI	Forecasting Comparison	Linear Regression	29 Features	Default	Validation	RMSE	5937.10	UCI_MODEL_COMPARISON.csv	Baseline Linear Model
UCI	Forecasting Comparison	Linear Regression	29 Features	Default	Validation	R2	0.991942	UCI_MODEL_COMPARISON.csv	Baseline Linear Model
UCI	Forecasting Comparison	Linear Regression	29 Features	Default	Validation	sMAPE	0.0200	UCI_MODEL_COMPARISON.csv	Baseline Linear Model
UCI	Forecasting Comparison	Ridge Regression	29 Features	Default	Validation	MAE	3994.50	UCI_MODEL_COMPARISON.csv	L2 Regularization
UCI	Forecasting Comparison	Ridge Regression	29 Features	Default	Validation	RMSE	5937.11	UCI_MODEL_COMPARISON.csv	L2 Regularization
UCI	Forecasting Comparison	Ridge Regression	29 Features	Default	Validation	R2	0.991942	UCI_MODEL_COMPARISON.csv	L2 Regularization
UCI	Forecasting Comparison	Decision Tree Regressor	29 Features	Default	Validation	MAE	6205.81	UCI_MODEL_COMPARISON.csv	Decision Tree
UCI	Forecasting Comparison	Decision Tree Regressor	29 Features	Default	Validation	RMSE	9070.78	UCI_MODEL_COMPARISON.csv	Decision Tree
UCI	Forecasting Comparison	Decision Tree Regressor	29 Features	Default	Validation	R2	0.981198	UCI_MODEL_COMPARISON.csv	Decision Tree
UCI	Forecasting Comparison	Random Forest Regressor	29 Features	Default	Validation	MAE	4297.09	UCI_MODEL_COMPARISON.csv	Random Forest Ensemble
UCI	Forecasting Comparison	Random Forest Regressor	29 Features	Default	Validation	RMSE	6320.15	UCI_MODEL_COMPARISON.csv	Random Forest Ensemble

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UCI	Forecasting Comparison	Random Forest Regressor	29 Features	Default	Validation	R2	0.990875	UCI_MODEL_COMPARISON.csv	Random Forest Ensemble
UCI	Forecasting Comparison	XGBoost Regressor	29 Features	Tuned	Validation	MAE	3935.63	UCI_MODEL_COMPARISON.csv	XGBoost Validation
UCI	Forecasting Comparison	XGBoost Regressor	29 Features	Tuned	Validation	RMSE	5802.40	UCI_MODEL_COMPARISON.csv	XGBoost Validation
UCI	Forecasting Comparison	XGBoost Regressor	29 Features	Tuned	Validation	R2	0.992300	UCI_MODEL_COMPARISON.csv	XGBoost Validation
UCI	Forecasting Comparison	XGBoost Regressor	29 Features	Tuned	Validation	sMAPE	0.0194	UCI_MODEL_COMPARISON.csv	XGBoost Validation
UCI	Time Series CV	Linear Regression	29 Features	Default	5-Fold Temporal CV	Mean MAE	8092.58	UCI_TIME_SERIES_CV.csv	Expanding window CV
UCI	Time Series CV	Ridge Regression	29 Features	Default	5-Fold Temporal CV	Mean MAE	8091.44	UCI_TIME_SERIES_CV.csv	Expanding window CV
UCI	Time Series CV	Random Forest Regressor	29 Features	Default	5-Fold Temporal CV	Mean MAE	8415.71	UCI_TIME_SERIES_CV.csv	Expanding window CV
UCI	Time Series CV	XGBoost Regressor	29 Features	Default	5-Fold Temporal CV	Mean MAE	10365.77	UCI_TIME_SERIES_CV.csv	Expanding window CV
UCI	Sealed Test Evaluation	XGBoost Regressor	29 Features	Frozen Model	Sealed Test	MAE	4421.55	UCI_ERROR_ANALYSIS.md	Final sealed test set (H2 2014)
UCI	Sealed Test Evaluation	XGBoost Regressor	29 Features	Frozen Model	Sealed Test	RMSE	6654.94	UCI_ERROR_ANALYSIS.md	Final sealed test set (H2 2014)
UCI	Sealed Test Evaluation	XGBoost Regressor	29 Features	Frozen Model	Sealed Test	R2	0.994400	UCI_ERROR_ANALYSIS.md	Final sealed test set (H2 2014)
UCI	Sealed Test Evaluation	XGBoost Regressor	29 Features	Frozen Model	Sealed Test	sMAPE	0.0193	UCI_ERROR_ANALYSIS.md	Final sealed test set (H2 2014)
UCI	Sealed Test Evaluation	Daily Seasonal Naive	1 Feature	Baseline	Sealed Test	MAE	20808.20	UCI_ERROR_ANALYSIS.md	Daily persistence benchmark